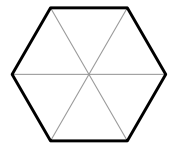
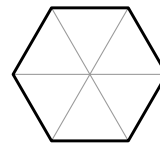
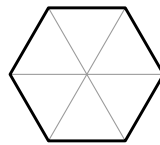
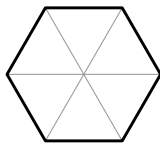
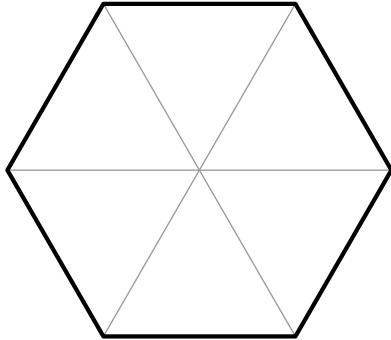


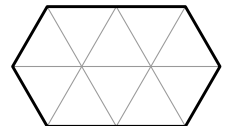
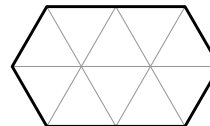
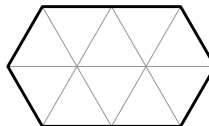
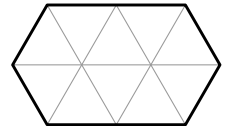
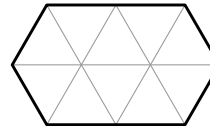
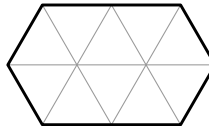
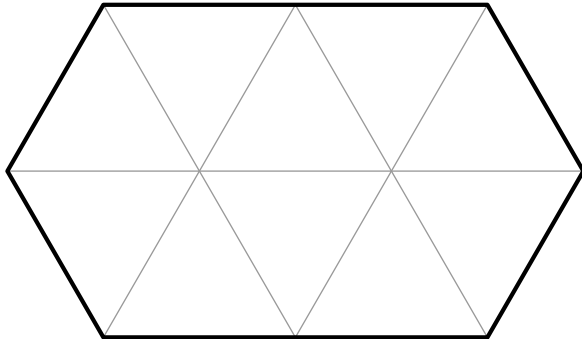
For every problem: use blue rhombuses only. To cover a board, fill all of it with no gaps, no overlaps, and nothing sticking out. Two coverings are different if their drawings are different, even when one is a turned or mirror copy of the other.

Problem 1: Find all the different ways to cover board A, and draw each one in the small pictures beside the board. Do the same for board B and board C. Then imagine a board shaped like these whose top edge is 10 small edges long. How many ways are there to cover it? Explain how you know.

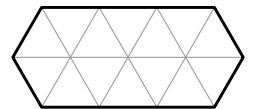
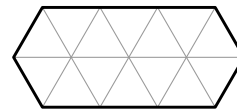
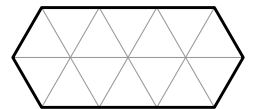
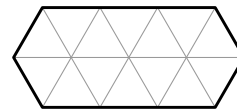
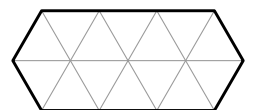
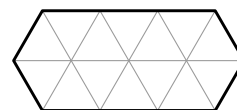
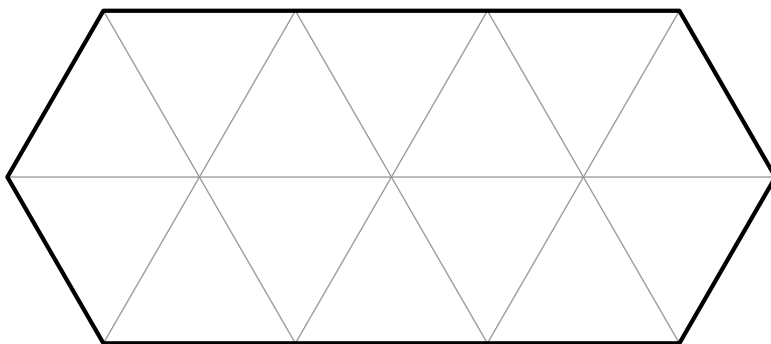
A



B



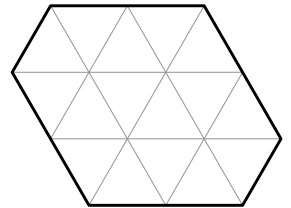
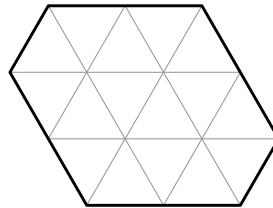
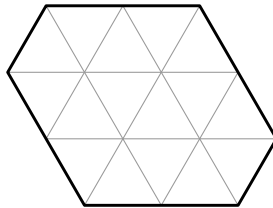
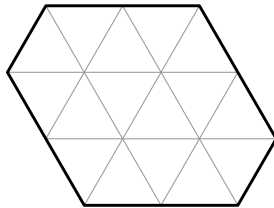
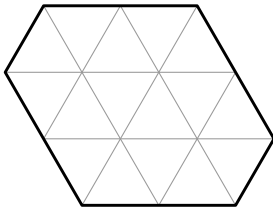
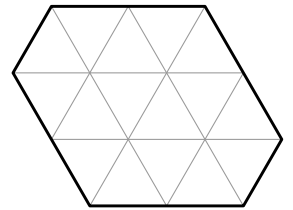
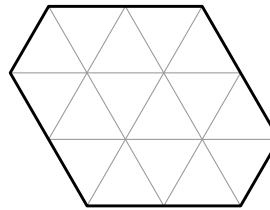
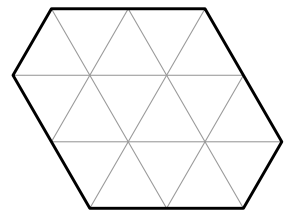
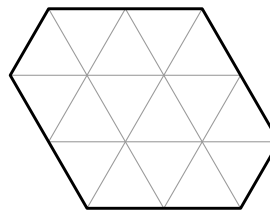
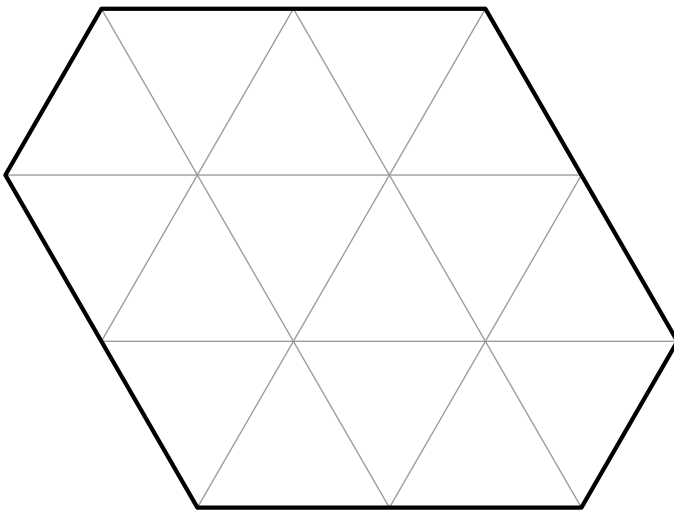
C



Board with a top edge 10 small edges long: _____ ways

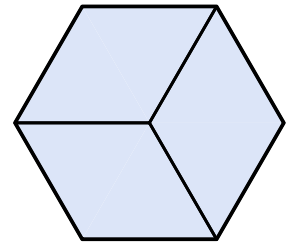
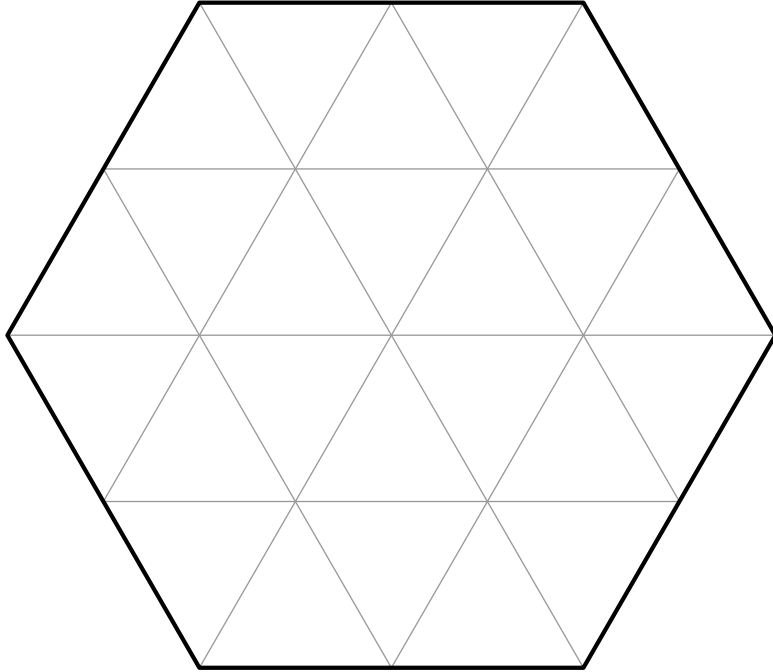
Problem 2: Find all the different ways to cover board D, and draw each one. Explain how you know you have found them all.

D

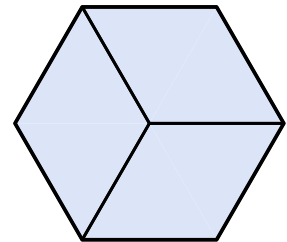


Problem 3: Cover board E to match picture S. Then change it into picture T, one move at a time. In one move, you find three rhombuses that together make the shape of a yellow hexagon, lift them out, and put them back the other way. What is the fewest moves that takes S to T? Do the same from S to U, and from U to T.

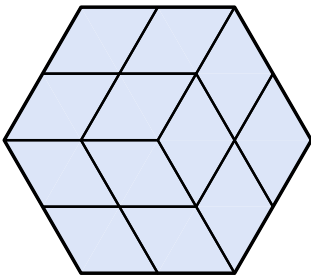
E



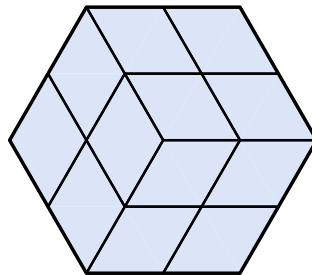
one move



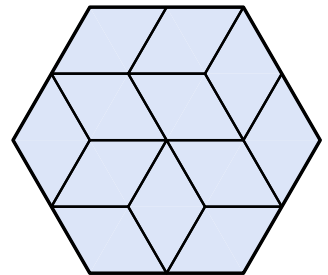
S



T



U

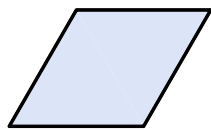


S to T: _____ moves

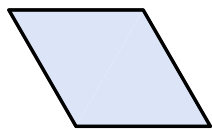
S to U: _____ moves

U to T: _____ moves

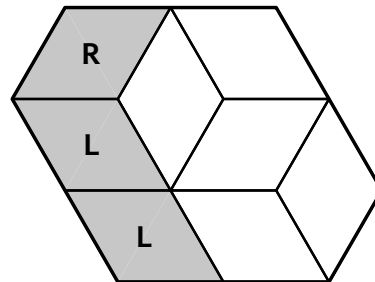
Problem 4: In a covering of a board, start at one small edge on the top side. Go down through the rhombus below it, out through that rhombus’s bottom edge, and into the next rhombus. Keep going until you reach the bottom side. Shade this chain of rhombuses, and write a letter for each rhombus in it, from top to bottom: R if it leans right, L if it leans left. The picture shows one chain in a covering of board D.



R (leans right)

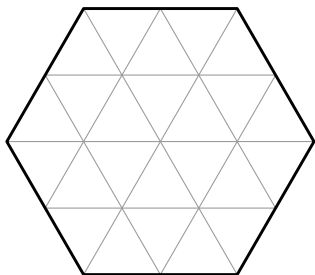


L (leans left)

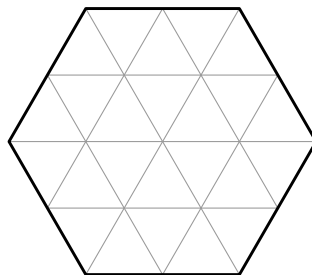


letters: RLL

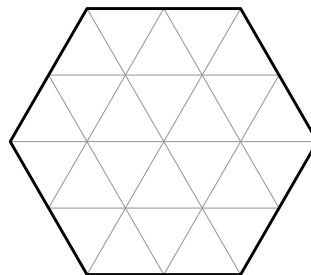
Shade both chains in each covering of board D that you drew in Problem 2, and write their letters. Then, for each pair below, draw the covering of board E whose left chain and right chain have these letters, or explain why there is none.



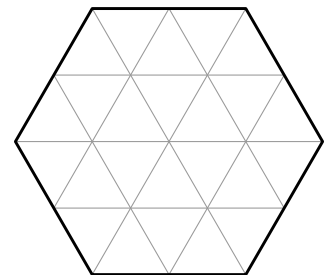
left: RLRL
right: RLLR



left: LRLR
right: RLRL

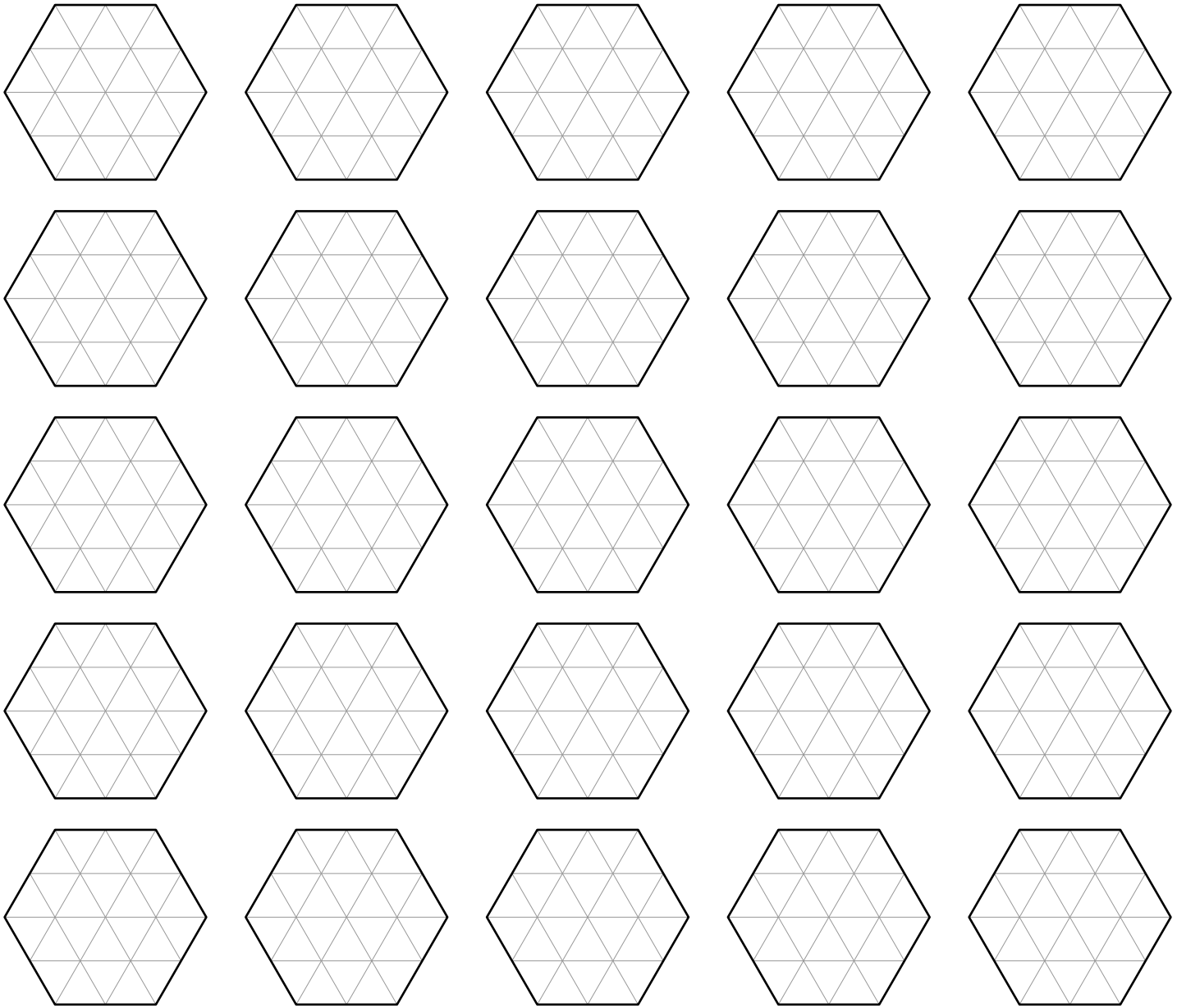


left: RRLL
right: LLRR



left: LRRL
right: RRLL

Problem 5: How many different ways are there to cover board E? Draw them. Explain how you know you have found them all.



Number of ways: _____

Problem 6: Cover board E any way you like, and write the letters of its two chains. Make one move, as in Problem 3, and write the letters again. Try this with many different moves. What does one move do to the letters?

Problem 7: Explain why it takes at least 8 moves to get from S to T. Then start with any covering of board E, make some moves, and finish with the same covering you started with. Can the number of moves be odd? Explain.
